

Thrombolytics for Cardiac Arrest: Case Report and Systematic Review of Controlled Trials

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The administration of systemic thrombolytic medications during cardiac arrest has been a topic of debate for many years. Adults have been successfully resuscitated following administration of a thrombolytic after initial failure of standard cardiopulmonary resuscitation (CPR) techniques, particularly when the condition leading to the arrest was acute pulmonary embolism (PE) or myocardial infarction.¹⁻⁷ Current Advanced Cardiac Life Support (ACLS) guidelines state that there is insufficient evidence to recommend for or against the routine use of thrombolytic drugs for cardiac arrest and it may be considered on a case-by-case basis when PE is suspected.^{8,9} We report the case of a 48-year-old man who survived following administration of tenecteplase during CPR for the presumptive diagnosis of PE. A review of the best available evidence for the use of thrombolysis during cardiac arrest is presented to further address this controversy.

Case Report

A 48-year-old male presented to the emergency department (ED) of a tertiary care, academic teaching hospital, with acute onset of shortness of breath, which began 2 hours prior to presentation. He

OBJECTIVE: To describe a successful case involving the use of tenecteplase during cardiac arrest for presumed pulmonary embolism (PE) and to systematically review the evidence from controlled trials supporting the efficacy and safety of thrombolysis during cardiac arrest.

CASE SUMMARY: A 48-year-old male presented to the emergency department with an acute onset of shortness of breath that began approximately 2 hours prior to presentation. Prior to undergoing a computed tomography (CT) scan to rule out PE, the patient went into cardiac arrest, with an initial rhythm of pulseless electrical activity at a rate of 140 beats/min. Cardiopulmonary resuscitation (CPR) was initiated and, due to suspected PE, a bolus dose of tenecteplase 50 mg was administered immediately following a single 1-mg dose of epinephrine. CPR was continued and 4 additional 1-mg doses of epinephrine and three 1-mg doses of atropine were given. After 13 minutes of CPR, return of spontaneous circulation (ROSC) was achieved, with a blood pressure of 144/50 mm Hg. After the patient was stabilized, a CT scan demonstrated extensive bilateral pulmonary emboli in most segmental arteries. He was admitted to the intensive care unit where he was sedated, paralyzed, and treated with induced hypothermia for 24 hours. He was discharged from the hospital 2 weeks later on warfarin, with no noted neurologic deficits.

DISCUSSION: A systematic search of MEDLINE (1950-August 2010), Embase (1980-August 2010), and Google Scholar (to August 2010) was conducted to identify prospective controlled trials that investigated the use of thrombolytic medications to treat cardiac arrest. Five trials involving 1544 undifferentiated cases of cardiac arrest were found. Overall, some trials reported an improved rate of ROSC following administration of thrombolytics, but there was no overall mortality reduction in any trial. There was, however, an increased risk of bleeding events following administration of a thrombolytic drug.

CONCLUSIONS: Controlled trials demonstrate that there is a lack of benefit and potential harm in administering thrombolysis in an undifferentiated patient with cardiac arrest. However, the case we present provides evidence that fibrinolysis may benefit selected patients with cardiac arrest in whom PE is confirmed or in whom there is high index of suspicion of PE.

KEY WORDS: cardiac arrest, fibrinolytics, pulmonary embolism, thrombolysis.

Ann Pharmacother 2010;44:2007-13.

Published Online, 30 Nov 2010, *theannals.com*, DOI 10.1345/aph.1P364

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had no associated chest pain, fever, or respiratory symptoms. Vital signs on presentation included heart rate 100 beats/min, blood pressure 112/88 mm Hg, and respiratory rate 16 breaths/min. Oxygen saturation on room air was 90%, which increased to 98% with 2 L oxygen via nasal prongs. There was no obvious distress at rest; however, after walking he became short of breath with an increased respiratory rate of 30 breaths/min. Results of a physical examination were unremarkable. The patient had a normal cardiorespiratory exam. He had prominent varicosities in his left lower extremity which had been present for many years. The initial electrocardiogram (ECG) revealed normal sinus rhythm with no signs of ischemia or ventricular strain (Figure 1). He denied use of any medications and his past medical history was significant for varicose veins and a PE approximately 15 years earlier following removal of a bladder hemangioma.

A computed tomography (CT) scan was ordered to rule out PE, and the patient was transferred to the CT scanner an hour after presentation. Upon arrival at the CT scanner, but prior to being transferred to the scanner table, he developed seizure-like activity. No imaging was performed, and he was immediately returned to the ED, where he was responsive only to deep sternal rub and developed central cyanosis. His peripheral pulses were evident only on Doppler scan, sinus rhythm at 136 beats/min was evident on the cardiac monitor, and his oxygen saturation level was unobtainable. While being prepared for intubation, the patient's heart rate decreased to 20 beats/min; he responded to atropine 1 mg, which in-

creased his heart rate to 88 beats/min. However, his level of consciousness deteriorated further and he became unresponsive, with a thready pulse. He was intubated without pharmacologic support and immediately thereafter developed narrow-complex, pulseless electrical activity (PEA) at a rate of 140 beats/min. CPR and ACLS protocols were initiated and a bolus dose of tenecteplase 50 mg was administered immediately following a single dose of epinephrine 1 mg. CPR was continued and 4 additional doses of epinephrine 1 mg and 3 more doses of atropine 1 mg were given. After 13 minutes of CPR, a perfusing rhythm was noted on the monitor, with blood pressure of 144/50 mm Hg. The patient remained unresponsive and postresuscitation blood pressure remained stable, with heart rate 120-125 beats/min. An ECG performed immediately postresuscitation displayed sinus tachycardia, a right bundle branch block, and right axis deviation (Figure 2). Eight hours following return of spontaneous circulation (ROSC), the ECG returned to normal (Figure 3). After he was stabilized, a CT scan demonstrated extensive bilateral pulmonary emboli in most segmental arteries. We hypothesize that this was due to thrombolysis of a saddle embolism during resuscitation. Cardiac septal deviation to the left was also noted on the CT scan and was presumed to be secondary to right heart strain. The patient was admitted to the ICU, where he was sedated, paralyzed, and treated with induced hypothermia for 24 hours. He was discharged from the hospital 2 weeks later on warfarin, with return to premonitory condition and no noted neurologic deficits.

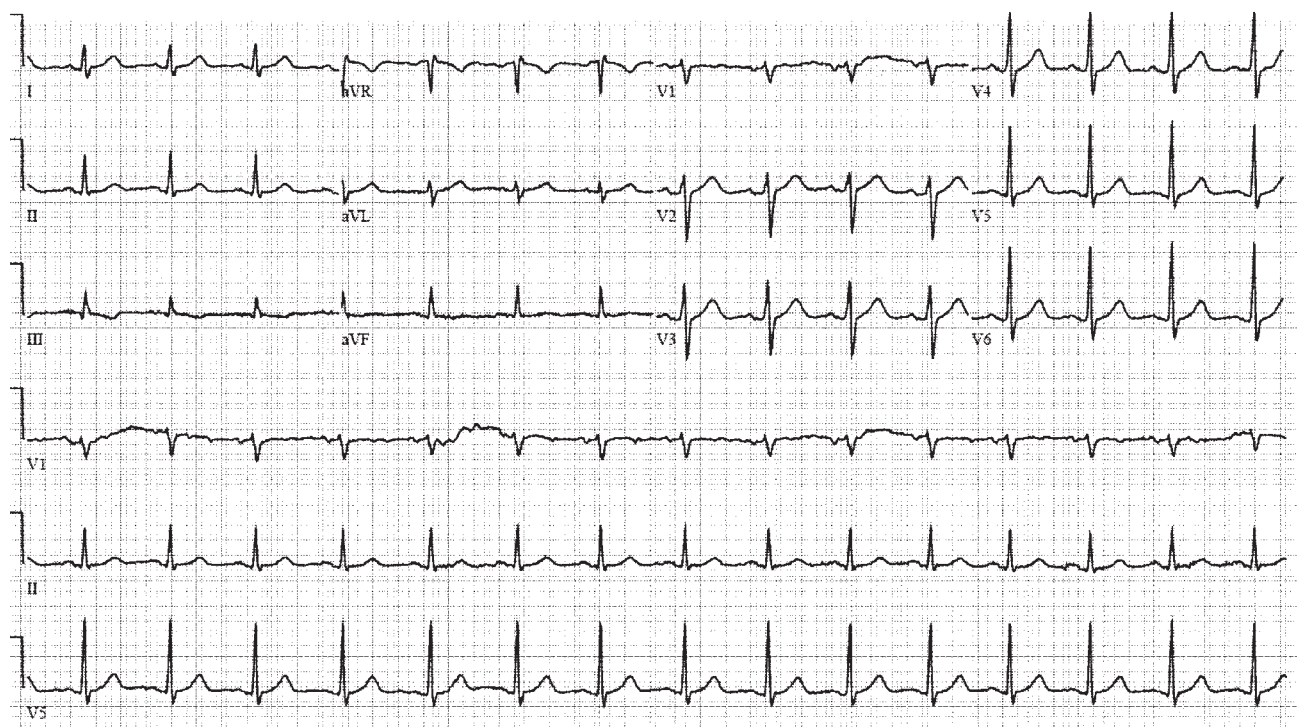


Figure 1. Initial electrocardiogram.

Discussion

PE is responsible for 200,000 deaths annually in the US, one third of which occur in the first hour from symptom onset.¹⁰ For most patients, anticoagulation with heparin or low-molecular-weight heparin (5-7 days) and warfarin is all that is required in the management of (noncardiac arrest) PE.¹¹ Among all patients presenting with nonmassive or submassive PE, thrombolysis does not offer any therapeutic advantage over anticoagulation.¹²⁻¹⁴ Although prospective randomized trials on thrombolysis in massive PE (hemodynamic instability with right ventricular dysfunction) are few, this therapy has become more common, as it has been shown to have an early effect on right ventricular function and pulmonary perfusion.^{15,16}

Cardiac arrest from PE can present with any rhythm; however, PEA is most commonly associated with this condition.^{10,17} PEA comprises 20% of cardiac arrests and has a survival to hospital discharge rate of approximately 4%.¹⁸ Most PEA survivors respond to initial interventions early during resuscitative efforts and, although poorly studied, this poor outcome might not be consistent across all subgroups, particularly patients with PE.^{18,19} PEA is unique among cardiac arrest rhythms in that forward blood flow can occur. This has prompted heightened vigilance for reversible causes when treating sustained PEA.²⁰ The possibility of low-grade cerebral and coronary perfusion during PEA supports the concept that a good outcome can result

even with prolonged cardiac arrest if such a cause is addressed. There are no validated criteria to guide clinicians in selecting a subgroup of cardiac arrest patients likely to have a PE, although it seems intuitive that risk factors for venous thromboembolic disease, particularly a previous event, as with our patient, would define a high-risk group.

Several case reports have described the use of thrombolysis during cardiac arrest.^{1-7,21-25} These reports have drawn from cases in which the known or highly suspected etiology of the arrest was thrombotic, either pulmonary or coronary. Given the variability and nature of resuscitation care and the potential for positive publication bias with case reports, randomized controlled trials are necessary to delineate the contribution that the administration of a thrombolytic had to the outcome. A systematic search of the literature was conducted to identify controlled clinical trials.

MEDLINE (1950-August 2010), Embase (1980-August 2010), and Google Scholar (to August 2010) were searched using the terms heart arrest, thrombolytic therapy, tissue plasminogen activator, alteplase, tenecteplase, clot lysis, and thrombolytic for cardiac arrest. Titles were screened by 1 author (JP) and relevant abstracts and full text articles were reviewed without the application of specific limits. Relevant information was extracted by 2 authors (JP, PJZ).

Five prospective, controlled trials involving 1544 patients have been published evaluating thrombolysis in cardiac arrest.²⁶⁻³⁰ Commonly, these trials used endpoints of



Figure 2. First electrocardiogram following return of spontaneous circulation.

ROSC and survival to various time points for efficacy, and bleeding for safety. A summary of the studies' efficacy results is presented in Table 1.

Böttiger et al.²⁶ performed a prospective observational study in 90 adults with nontraumatic cardiac arrest who failed to achieve ROSC after 15 minutes of standard ACLS care. More than half of the patients had asystole as their initial rhythm, with one third experiencing ventricular fibrillation (VF)/pulseless ventricular tachycardia (pVT). Control patients were enrolled for 1 year, all of whom received standard resuscitation care; the following year, the intervention group received the same standard resuscitation care plus tissue plasminogen activator (tPA) 50 mg intravenous push and heparin 5000 units intravenous push. Doses were potentially repeated after an additional 30 minutes of CPR in the absence of ROSC. Despite the lack of randomization, patient baseline characteristics were well matched. The study showed a significant increase in rates of ROSC and survival to ICU admission, but failed to show significant improvements in overall survival. Two bleeding events were reported, both in the tPA group (on days 2 and 12 postadministration of tPA).

This small study was the first published controlled trial that demonstrated a possible benefit to administering thrombolytic medication to patients in cardiac arrest. Enrollment of all patients who had cardiac arrest versus just those with suspected thrombotic etiology possibly reduced the study's power to detect a benefit; however, this study served as a foundation for further research. From a safety

perspective, the low number of events precludes any meaningful statistical analysis.

Abu-Laban et al.²⁷ conducted a prospective, randomized, double-blind, placebo-controlled study that enrolled out-of-hospital cardiac arrest patients in PEA arrest. Their population was similar to that of the previous study, with a mean age of 69-70 years and a majority (~70%) of male patients. The initial rhythm was PEA in approximately 50%, VF/pVT in approximately 25%, and asystole in approximately 25% of patients. Additionally, at least 1 minute of PEA arrest was a criterion for enrollment. Prior to enrollment and randomization, all patients received NaCl 0.9% 500 mL and epinephrine 1 mg. Patients were randomized to receive intravenous tPA 100 mg over 15 minutes or placebo during their resuscitation by emergency medical services (ambulance) personnel, and CPR was continued for at least 15 minutes after study drug administration. Of 233 patients enrolled, only 1 (in the tPA arm) survived to hospital discharge. Two major bleeds occurred during the study, both in the tPA-treated group.

This was the first randomized controlled trial on the topic whose results were published. Although the reason was not described within the study text, enrollment of patients with PEA arrest was likely intended as a means of selecting a population that would potentially be more likely to respond to pharmacologic therapy; that being fewer patients with VF/pVT where defibrillation is the best intervention and asystole where survival rates are extremely poor. That they were not able to specifically select patients with likely throm-

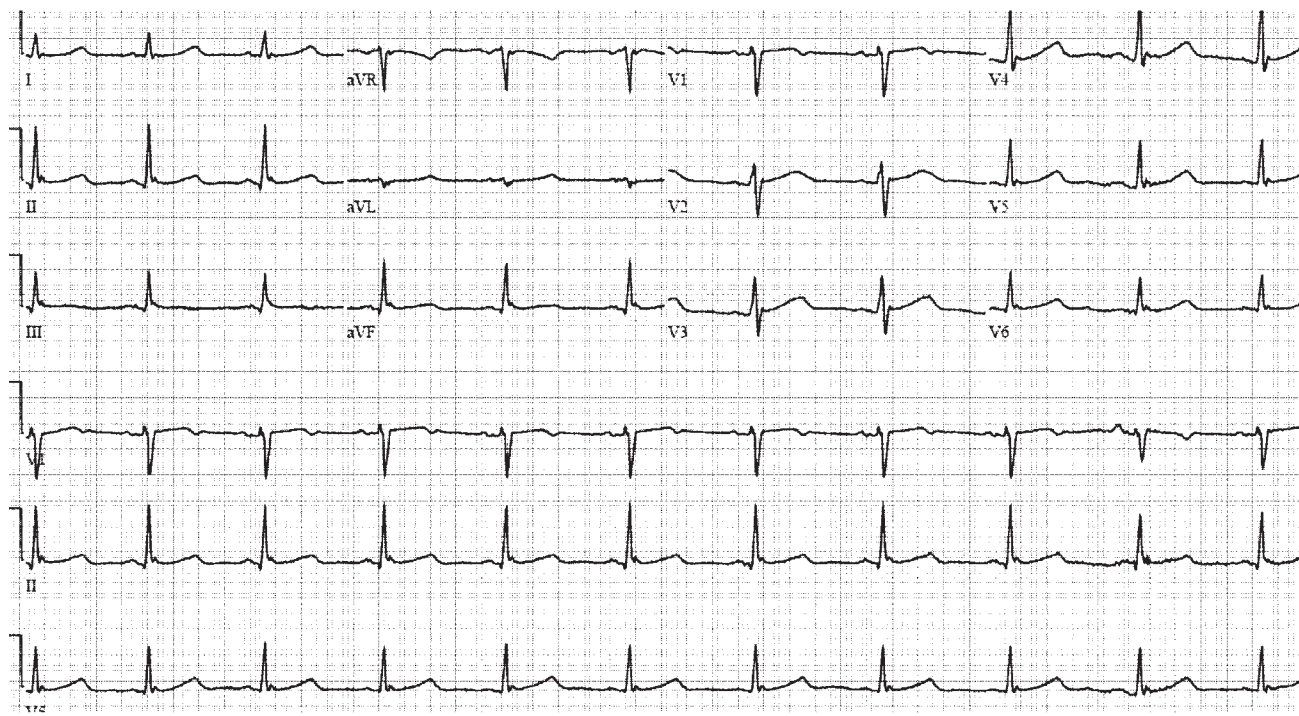


Figure 3. Electrocardiogram eight hours following return of spontaneous circulation.

Table 1. Study Characteristics and Outcomes for Thrombolytic Drugs During Cardiac Arrest

Reference	ROSC, n/N (%)		Survival to ED Discharge or ICU/Hospital Admission, n/N (%)		Survival 24 Hours, n/N (%)		Survival to ICU Discharge, n/N (%)		Survival to Hospital Discharge, n/N (%)	
	Lytic	Control	Lytic	Control	Lytic	Control	Lytic	Control	Lytic	Control
Böttiger (2001) ²⁶	27/40 (68) ^a	22/50 (44)	23/40 (58) ^a	15/50 (30)	14/40 (35)	11/50 (22)	NR	NR	6/40 (15)	4/50 (8)
Abu-Laban (2002) ²⁷	25/117 (21)	27/116 (23)	7/117 (6)	6/116 (5)	NR	NR	NR	NR	1/117 (1)	0/116 (0)
Fatovich (2004) ²⁸	8/19 (42) ^a	1/16 (6)	2/19 (10)	1/16 (6)	NR	NR	1/19 (5)	1/16 (6)	1/19 (5)	1/16 (6)
Bozeman (2006) ²⁹	13/50 (26) ^a	14/113 (12)	6/50 (12) ^a	0/113 (0)	2/50 (4)	0/113 (0)	NR	NR	2/50 (4)	0/113 (0)
Böttiger (2008) ³⁰	283/515 (55)	279/511 (55)	281/525 (54)	289/525 (55)	158/517 (31)	171/514 (33)	NR	NR	77/525 (15)	89/525 (18)

ED = emergency department; ICU = intensive care unit; Lytic = thrombolytic drug; NR = not reported; ROSC = return of spontaneous circulation.

^aStatistically significant ($p < 0.05$) versus respective control.

botic etiology of cardiac arrest, again, may have diluted the investigators' effectiveness outcomes. The authors go so far as to suggest that further study in that selected population may be of benefit. The true randomized design of this study does increase the internal validity of the results.

The TICA (Thrombolysis in Cardiac Arrest) trial results, published by Fatovich et al.,²⁸ described a prospective, randomized, double-blind, placebo-controlled pilot study that enrolled 35 patients with out-of-hospital cardiac arrest (any rhythm). Patients were randomized to receive tenecteplase 50 mg intravenous push or placebo plus standard care upon arrival at the hospital. There were baseline imbalances between the tenecteplase and placebo groups in terms of sex (58% vs 75% male), mean age (63 years vs 72 years), location of arrest (32% vs 12% public place), and initial rhythm (63% vs 19% VF/pVT and 37% vs 75% PEA), respectively. In the country in which the study was conducted, emergency medical personnel do not administer intravenous medications, so the study was designed to give tenecteplase as the first pharmacologic intervention during the resuscitation period. There was an increased rate of ROSC with administration of a thrombolytic but no difference in mortality. No bleeding complications were reported. Autopsy information was provided for 4 patients in each group, with 2 in the tenecteplase group and 3 in the placebo group demonstrating a thrombotic (myocardial infarction) cause of death.

This small trial unfortunately adds very little to the previous study, as its small sample introduced significant confounding from the baseline characteristics of the patients. Additionally, it is potentially difficult to extrapolate the results to clinical practice in North America, where ambulance personnel are qualified to administer a full complement of intravenous ACLS resuscitation medications.

Bozeman et al.²⁹ published a prospective cohort study in patients with nontraumatic cardiac arrest refractory to standard ACLS guidelines. Fifty patients were enrolled in an intervention arm and 113 patients were enrolled as controls. The intervention group received tenecteplase 0.5 mg/kg intravenous push while controls were concurrently recruited from patients who did not respond to standard ACLS treatment but did not receive thrombolysis. At baseline, the tenecteplase group was younger (mean age 58 years vs 65 years), had different arrest locations (80% vs 68% out-of-hospital), had more witnessed arrests (82% vs 54%), and had different initial rhythm rates (28% vs 20% VF; 32% vs 38% PEA; 40% vs 39% asystole) compared with the control group. In addition to different study drugs, management of these patients was also imbalanced, with 64% versus 47% of tenecteplase and control patients, respectively, being defibrillated. Of the 163 patients enrolled, only 2 survived to hospital discharge, both in the tenecteplase group, while 1 patient in the tenecteplase group had a subarachnoid hemorrhage.

The results of this study again demonstrated the poor prognosis associated with cardiac arrest and did not demonstrate a clinical benefit for thrombolysis. Methodologically, the use of an unmatched, nonrandomized cohort design makes it difficult to attempt to determine the contribution of any particular intervention. Outcomes were likely skewed by selection bias and other unspecified differences in the care of the patients.

The TROICA (Thrombolysis Using Tenecteplase in Cardiac Arrest) study, by Bottiger et al.,³⁰ is the largest randomized, double-blind, placebo-controlled trial published to date; 1050 patients with out-of-hospital cardiac arrest, who remained in arrest after 10 minutes of CPR, were randomly assigned to receive tenecteplase 0.5 mg/kg intravenous push or placebo. Patients were well matched between groups, with a mean age of 65 years, approximately 80% male, time to first CPR (9 minutes), time to study drug administration (18 minutes), and initial rhythm (55% VF, 22% PEA, 22% asystole). Management with other interventions, including ACLS medications, aspirin, and heparin, was also well matched between groups. There were differences regarding presumed or proven etiology of the cardiac arrest between the tenecteplase and placebo groups with myocardial infarction (75% vs 69%) and PE (6% vs 11%). Despite protocol modifications midtrial, to correct for low survival rates, the study failed to demonstrate any significant benefits in any clinical efficacy outcomes, including ROSC and survival, and was stopped early after a futility analysis. The investigators did, however, demonstrate a significant increase in the rate of intracranial hemorrhage with the use of tenecteplase (2.7% vs 0.4%, absolute risk increase 2.3%, number needed to harm 43; $p = 0.006$). Despite numerous post hoc analyses to determine patient characteristics that might favor tenecteplase, only bystander CPR (with or without an initial rhythm of VF) was associated with better outcomes with tenecteplase treatment.

The key issue that potentially contributed to this study's negative outcome was the exclusion of patients who were presumed to have PE as the cause of cardiac arrest. These patients were given open-label thrombolytics and were excluded from enrollment. One can postulate that, had they been included, these patients may have driven the effectiveness and perhaps improved outcomes in the tenecteplase group. Improved outcome in patients with early bystander CPR was more beneficial in the tenecteplase group, but could also be attributed to statistical chance, as only 2 of the 24 presented post hoc analyses reached statistical significance at a p value of 0.05 (not corrected for multiple analyses). The statistically significant increase in serious bleeding events should provide further argument against providing thrombolysis to all patients with cardiac arrest.

Overall, evidence from these controlled trials for the use of thrombolysis in undifferentiated cardiac arrest is unin-

spiring. However, other literature, primarily case reports, as well as the case we present, does provide a compelling argument for a potential benefit of thrombolysis in cardiac arrest when PE is known or highly suspected, and certainly supports the concept that complete recovery can result despite prolonged cardiac arrest. Unfortunately, it is impossible to know how many times this therapy has been administered unsuccessfully and what impact reporting bias may have played in the predominately positive published cases seen with case reports.

The challenge in the ED, when confronted with a patient in whom the diagnosis of PE-induced cardiac arrest is certain or highly suspected, is that it is unclear from the literature whether thrombolytic administration may be beneficial. From a clinical decision-making perspective, controlled trials have shown that there is a lack of benefit and potential harm to administering thrombolysis in a patient with undifferentiated cardiac arrest; however, the case we present provides further evidence to previous reports that thrombolysis may benefit select patients with cardiac arrest in whom PE is confirmed or there is a high index of suspicion. Clinicians should be cognizant of current evidence and not "write off" a potentially beneficial intervention in a patient who is part of a population that has not been adequately studied and who likely has a better than average chance of benefiting from this therapy.

Therefore, we suggest that thrombolysis should be considered in patients who are known or highly suspected to have PE-induced cardiac arrest, particularly if other encouraging signs are present such as narrow QRS complexes (particularly if the arrest rhythm is PEA), significant levels of end-tidal CO_2 , evidence of forward blood flow by Doppler scan or invasive monitoring, or evidence of cardiac activity by echocardiography. If used, simple to administer regimens that have been shown to be feasible in trials, such as intravenous tPA 100 mg over 15 minutes or intravenous bolus tenecteplase 0.5 mg/kg (50 mg maximum), can be considered (although other regimens may also be appropriate). Following administration, CPR should continue for at least 15 minutes to allow sufficient time for the medication to act.

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Reprints/Online Access: www.theannals.com/cgi/reprint/aph.1P364

Conflict of interest: Authors reported none

References

- Bailen MR, Cuadra JAR, de Hoyos EA. Thrombolysis during cardiopulmonary resuscitation in fulminant pulmonary embolism: a review. *Crit Care Med* 2001;29:2011-9.
- Zahorec R. Rescue systemic thrombolysis during cardiopulmonary resuscitation. *Bratisl Lek Listy* 2002;103:266-9.
- Adams BD, Kim JY, Jackson WO. Tenecteplase and return of spontaneous circulation after refractory cardiopulmonary arrest. *South Med J* 2004;97:1015-7.
- Bauer MP, Vliegen HW, Huisman MV. Massive pulmonary embolism with cardiac arrest after intracardiac electrophysiological study: a strong case for venous thromboprophylaxis. *Blood Coagul Fibrinolysis* 2006;17:57-8.
- Hefer DVF, Munir A, Khouli H. Low-dose tenecteplase during cardiopulmonary resuscitation due to massive pulmonary embolism: a case report and review of previously reported cases. *Blood Coagul Fibrinolysis* 2007;18:691-4.
- Flores J, de Tena JG, de Pablo R, Daguerre M. Successful outcome of cardiopulmonary arrest with systemic thrombolysis. *Eur J Intern Med* 2008;19:e38-9. DOI 10.1016/j.ejim.2008.01.015
- Lapostolle F, Levasseur J, Dardel N, et al. Cardiac arrest after air travel successfully treated by presumptive fibrinolysis. *Resuscitation* 2009;80:606. DOI 10.1016/j.resuscitation.2009.02.010
- 2005 American Heart Association Guidelines for Cardiopulmonary Resuscitation and Emergency Cardiovascular Care. Part 7.2: Management of cardiac arrest. *Circulation* 2005;112(suppl I):IV58-66. DOI 10.1161/circulationaha.166557
- 2005 American Heart Association Guidelines for Cardiopulmonary Resuscitation and Emergency Cardiovascular Care. Part 7.4: Monitoring and medications. *Circulation* 2005;112(suppl I):IV78-83. DOI 10.1161/circulationaha.166559
- Kurkciyan I, Meron G, Sterz F, et al. Pulmonary embolism as cause of cardiac arrest: presentation and outcome. *Arch Intern Med* 2000;160:1529-35.
- Kearon C, Kahn SR, Agnelli G, Goldhaber S, Raskob GE, Comerota AJ. Antithrombotic therapy for venous thromboembolic disease: American College of Chest Physicians evidence-based clinical practice guidelines. 8th ed. *Chest* 2008;133:454S-545S. DOI 10.1378/chest.08-0658
- Dong BR, Hao Q, Yue J, Wu T, Liu GJ. Thrombolytic therapy for pulmonary embolism. *Cochrane Database of Systematic Reviews* 2009;3:CD004437. DOI 10.1002/14651858.CD004437.pub3
- Worster A, Smith C, Silver S, Brown MD. Thrombolytic therapy for sub-massive pulmonary embolism? *Ann Emerg Med* 2007;50:78-84. DOI 10.1016/j.annemergmed.2007.01.002
- Konstantinides S, Geibel A, Heusel G, Heinrich R, Kasper W. Heparin plus alteplase compared with heparin alone in patients with submassive pulmonary embolism. *N Engl J Med* 2002;347:1143-50.
- Jerjes-Sanches C, Ramirez-Rivera A, de Lourdes Garcia M, et al. Streptokinase and heparin versus heparin alone in massive pulmonary embolism: a randomized controlled trial. *J Thromb Thrombolysis* 1995;2:227-9. DOI 10.1007/bf01062714
- Goldhaber SZ, Haire WD, Feldstein ML, et al. Alteplase versus heparin in acute pulmonary embolism: randomized trial assessing right-ventricular function and pulmonary perfusion. *Lancet* 1993;341:507-11.
- Courtney DM, Sasser HC, Pincus CL, Kline JA. Pulseless electrical activity with witnessed arrest as a predictor of sudden death from massive pulmonary embolism in outpatients. *Resuscitation* 2001;49:265-72.
- Stueven HA, Aufderheide T, Waite EM, Mateer JR. Electromechanical dissociation: six years prehospital experience. *Resuscitation* 1989;17:173-82.
- Bonnin MJ, Pepe PE, Kimball KT, Clark PS. Distinct criteria for termination of resuscitation in the out-of-hospital setting. *JAMA* 1993;270:1457-62.
- Cripps T, Camm J. The management of electromechanical dissociation. *Resuscitation* 1991;22:173-80.
- Langdon RW, Swicegood WR, Schwartz DA. Thrombolytic therapy of massive pulmonary embolism during prolonged cardiac arrest using recombinant tissue-type plasminogen activator. *Ann Emerg Med* 1989;18:678-80.
- Tiffany PA, Schultz M, Steuven H. Bolus thrombolytic infusions during CPR for patients with refractory arrest rhythms: outcome of a case series. *Ann Emerg Med* 1998;31:124-6. DOI 10.1016/s0196.0644(98)70295-1
- Newman DH, Greenwald I, Callaway CW. Cardiac arrest and the role of thrombolytic agents. *Ann Emerg Med* 2000;35:472-80. DOI 10.1067/mem.2000.105600
- Lederer W, Lichtenberger C, Pechlaner C, Kroesen G, Baubin M. Recombinant tissue plasminogen activator during cardiopulmonary resuscitation in 108 patients with out-of-hospital cardiac arrest. *Resuscitation* 2001;50:71-6. DOI 10.1016/S0300-9572(01)00317-3
- Clark K, Abu-Laban RB, Zed PJ, Graham L. Neurologically normal survival after fibrinolysis during prolonged cardiac arrest: case report and discussion. *Can J Emerg Med* 2003;5:49-53.
- Bottiger BW, Bode C, Kern S, et al. Efficacy and safety of thrombolytic therapy after initially unsuccessful cardiopulmonary resuscitation: a prospective clinical trial. *Lancet* 2001;357:1583-5. DOI 10.1016/s0140-6736(00)04726-7
- Abu-Laban RB, Christenson JM, Innes GD, et al. Tissue plasminogen activator in cardiac arrest with pulseless electrical activity. *N Engl J Med* 2002;346:1522-8.
- Fatovich DM, Dobb GJ, Clugston RA. A pilot trial of thrombolysis in cardiac arrest (the TICA trial). *Resuscitation* 2004;61:309-13. DOI 10.1016/j.resuscitation.2004.01.016
- Bozeman WP, Kleiner DM, Ferguson KL. Empiric tenecteplase is associated with increased return of spontaneous circulation and short term survival in cardiac arrest patients unresponsive to standard interventions. *Resuscitation* 2006;69:399-406. DOI 10.1016/j.resuscitation.2005.09.027
- Bottiger BW, Arntz HR, Chamberlain DA, et al. Thrombolysis during resuscitation for out-of-hospital cardiac arrest. *N Engl J Med* 2008;359:2651-62.